

Internship Summary

YWL Engineering

Reflections from Weeks 1–10

Key Roles & Responsibilities

-  Develop and scale infrastructure systems
-  Design & implement full-stack modules: CRM, file manager, chat, calendar
-  Manage Ubuntu + Windows systems, VMs, and remote endpoints
-  Conduct vulnerability assessments & monitoring
-  Document workflows, deployments, and automation scripts
-  Collaborate across development and operations teams

Daily Operation Tasks

-  Write automation scripts (Python, Bash, PowerShell)
-  Prisma + PostgreSQL: database modeling and migrations
-  Hardware setups: system installations, replacements
-  Troubleshooting: software bugs, user requests, system downtimes
-  Sync with external GitLab repos & global dev branches
-  Maintain clear operational documentation and change logs

Key Achievements

-  Designed role/location-based access system
-  Built real-time file manager with drag & drop & search
-  Integrated AI embedding model with pgvector search
-  Built custom form system with live response viewer
-  Detected & resolved traversal and auth bypass risks
-  Led data recovery efforts after unexpected hardware failure
-  Developed workflow calendar and dynamic dashboards

Challenges & Solutions

 Challenge	 Solution
SMTP delivery failures	Built alert popups & admin control board
Validation/data sync errors	Improved frontend schema checks & feedback flow
IP address conflicts	Implemented ARP scan with subnet hint logic
Model overload	Refactored AI pipelines with job queueing
Windows script unreliability	Migrated to .bat/VBA fallback + log handlers
Disk corruption	Recovered with image cloning + backup sync

Knowledge Gained

-  **Infra & Networking:** Active Directory, Firewall, VPN, SMB
-  **AI Search:** BLIP captioning, cosine similarity, vector DBs
-  **DevOps:** Microservices, RESTful APIs, Prisma schema best practices
-  **Frameworks:** Next.js, Flask, ShadCN, Server-Sent Events (SSE)

Skills Acquired

-  Full-stack development with modular clean architecture
-  Cross-platform scripting and troubleshooting
-  Real-time monitoring & proactive alerting
-  UI/UX enhancements with feedback loops
-  System repair and rapid hardware redeployment

Attitudes Developed

-  Solution-focused mindset during fast-paced incidents
-  Code hygiene and commit discipline
-  Resilience under pressure and uncertainty
-  Collaboration across diverse technical stacks
-  Curiosity-driven learning and initiative-taking

Proposed Future Enhancements

-  Containerize AI & monitoring services for scalability
-  Deploy audit log with user tracing and tamper checks
-  Add version control and preview support for uploads
-  Expand resource & security dashboards (CPU, auth logs)
-  Rewrite Windows installers with verbose logging
-  Gantt chart for task dependencies and workflow planning
-  Integrate firewall logs with real-time analysis

 Thank You!

“Not just code — but growth, grit, and gratitude.”

Internship at YWL Engineering, Weeks 1–10